

# Computer Forensics

## 1. Overview of Computer Forensics

Computer Forensics is the process of collecting, preserving, analyzing, and presenting digital evidence in a legally acceptable manner.

### Objectives

- Identify cyber crimes
- Recover deleted data
- Find evidence
- Present evidence in court

### Examples

- Email investigation
  - Hacking investigation
  - Mobile forensics
  - Network forensics
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## 2. Difference Between Computer Crime and Unauthorized Activities

| Computer Crime                     | Unauthorized Activity             |
|------------------------------------|-----------------------------------|
| Illegal activity punishable by law | Violation of company policy       |
| Hacking bank server                | Using office PC for personal work |
| Data theft                         | Installing unauthorized software  |
| Legal action possible              | Administrative action possible    |

### Example

- Stealing customer data   Computer Crime
  - Watching YouTube on office PC   Unauthorized Activity
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### 3. Cyber Laws

Cyber laws govern activities performed using computers and networks.

#### In India

Main law:

Information Technology Act 2000

#### Important Sections

| Section | Purpose                    |
|---------|----------------------------|
| 43      | Unauthorized access        |
| 66      | Hacking                    |
| 66C     | Identity theft             |
| 66D     | Online fraud               |
| 67      | Publishing obscene content |

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## 4. Process of Computer Forensics (Six Steps)

### 1. Identification

Identify digital evidence.

### 2. Preservation

Protect evidence from modification.

### 3. Collection

Acquire evidence.

### 4. Examination

Extract useful information.

### 5. Analysis

Interpret evidence.

### 6. Reporting

Prepare final report.

### Mnemonic

I P C E A R

I - Identification

P - Preservation

C - Collection

E - Examination

A - Analysis

R - Reporting

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## 5. Need for Forensics Investigator

A forensic investigator is needed to:

- Investigate cyber crimes
- Recover deleted files
- Analyze malware
- Collect legal evidence
- Support law enforcement
- Prepare court reports

## Skills Required

- Networking
  - Operating Systems
  - Security
  - Legal Knowledge
  - Documentation
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## 6. Computer Forensics Involves

### Preservation

Protect evidence from damage.

### Identification

Locate relevant evidence.

### Extraction

Recover data from storage.

### Documentation

Record every activity.

### Interpretation

Analyze findings and draw conclusions.

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## 7. Goals of Forensics Analysis

### Main Goals

1. Determine what happened.
2. Identify attacker.
3. Recover evidence.
4. Preserve evidence integrity.

5. Support legal proceedings.

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## **8. Types of Cyber Forensics Techniques**

### **Disk Forensics**

Analysis of hard drives and SSDs.

### **Network Forensics**

Analysis of network traffic.

### **Mobile Forensics**

Investigation of smartphones.

### **Memory Forensics**

Analysis of RAM data.

### **Database Forensics**

Investigation of databases.

### **Malware Forensics**

Analysis of malicious software.

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## **9. Cyber Forensics Procedures**

### **Step 1**

Preparation

### **Step 2**

Detection

### **Step 3**

Containment

### **Step 4**

Collection

### **Step 5**

Analysis

### **Step 6**

Reporting

### **Step 7**

Recovery

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## **10. Preparation (Before Incident)**

Preparation is the most important phase.

### **Activities**

- Create incident response plan
  - Train employees
  - Backup critical data
  - Install security tools
  - Define response procedures
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## **11. What To Do Before the Incident**

### **Checklist**

Backup data

Maintain logs

Configure monitoring

Define team responsibilities

Create emergency contacts

Prepare forensic tools

Security awareness training

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## 12. Incident Response Plan (IRP)

An Incident Response Plan is a documented procedure for handling security incidents.

### Phases

Preparation

Identification

Containment

Eradication

Recovery

Lessons Learned

### Benefits

- Faster response
  - Reduced damage
  - Better evidence handling
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## 13. Incident Response Team (IRT)

A group responsible for handling security incidents.

### Members

- Security Analyst
- Forensic Investigator
- System Administrator
- Network Administrator
- Legal Team
- Management

### Responsibilities

- Detect incidents
  - Collect evidence
  - Investigate attacks
  - Restore services
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## 14. Detecting Incidents

### Common Indicators

- Unusual login attempts
- High CPU usage
- Unknown processes
- Malware alerts
- Suspicious network traffic
- Unauthorized file changes

### Detection Tools

- Wireshark
- Snort
- Suricata
- ELK Stack
- OSSEC



- Nagios
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## 15. Chain of Custody (Very Important)

### Definition

Chain of Custody is the documented history of evidence from collection to presentation in court.

### Purpose

- Maintain integrity
- Prevent tampering
- Ensure legal admissibility

### Information Recorded

- Evidence ID
- Date & Time
- Collector Name
- Transfer Details
- Purpose
- Signatures

### Example

Laptop Seized

Collected by Investigator A

Transferred to Lab Analyst

Analyzed

Stored Securely

Presented in Court

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# Most Important Exam Questions

1. Define Computer Forensics.
2. Difference between Computer Crime and Unauthorized Activity.
3. Explain Cyber Laws.
4. Explain the six steps of Computer Forensics.
5. What is the role of a Forensics Investigator?
6. Explain Preservation and Identification.
7. What are the goals of Forensics Analysis?
8. Types of Cyber Forensics Techniques.
9. Explain Incident Response Plan.
10. Explain Incident Response Team.
11. What is Chain of Custody?
12. Why is Chain of Custody important?
13. What should be done before a security incident?
14. Explain incident detection methods.
15. Explain the cyber forensics procedure.